

Learning Objectives

1. Determine the dependence of concentration on time elapsed during a chemical reaction using plots of the Integrated Rate Laws for First, Second, and Zero Order kinetics.
2. Calculate the time required for the reactant concentration to decrease a specified amount for First Order reactions.
3. Calculate the concentration of the reactant after the reaction has progressed a specified time for First Order reactions.
4. Understand the concept of half-life.
5. Calculate the rate constant for First Order reactions given the half-life for the reaction.
6. Distinguish between First, Second, and Zero Order kinetics by examining the dependence of half-life on concentration.

Integrated Rate Law

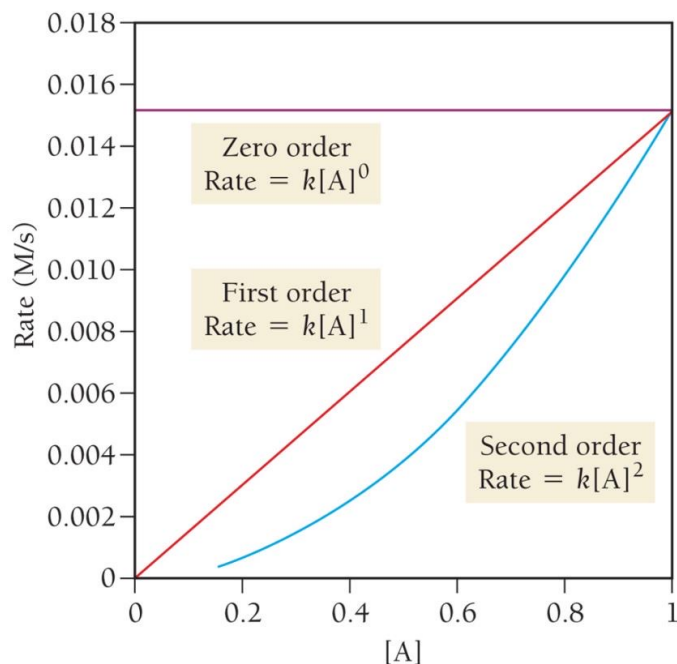
Rate Law:

- Zero order: $r = k$
- First order: $r = k[A]$
- Second order: $r = k[A]^2$

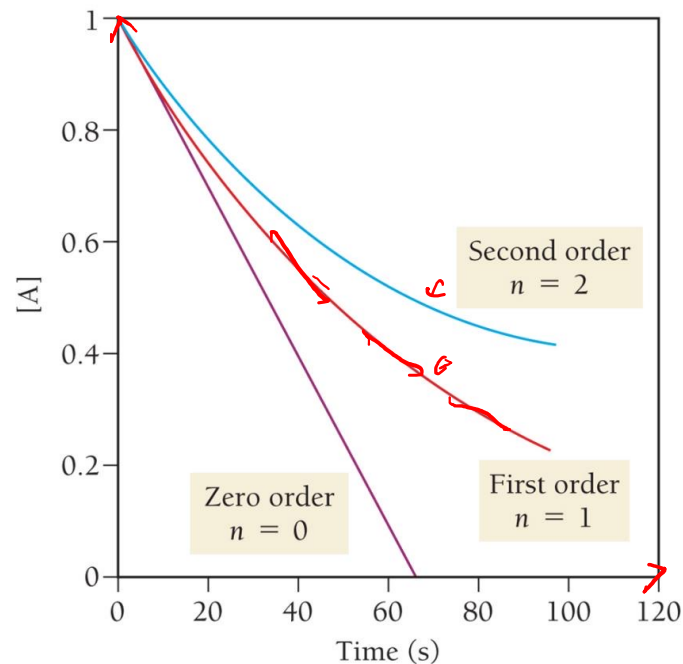
The integrated Rate Law:

- Describes the relationship between concentration and time

Rate versus Reactant Concentration



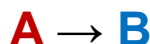
Reactant Concentration versus Time



Integrated Rate Law

The integrated Rate Law:

- Describes the relationship between concentration and time
- Derived from the Rate Law.



$$r = \frac{-d[A]}{dt}$$

Zero order reaction:

$$r = k = \frac{-d[A]}{dt}$$

$$-d[A] = k dt$$

$$d[A] = -k dt$$

$$\int_{t=0}^{t=t} d[A] = \int_{t=0}^{t=t} -k dt$$

$$[A] \Big|_0^t = -kt \Big|_0^t$$

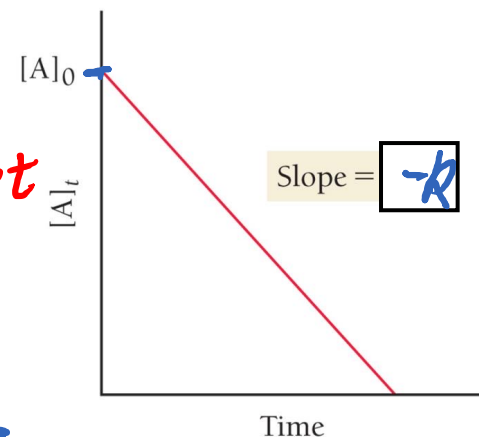
$$[A]_t - [A]_0 = -k(t - 0) = -kt$$

$$[A]_t = [A]_0 - kt$$

$$y = m - kt$$

- Integrated Rate Law for zero order reaction:

$$[A]_t = [A]_0 - kt$$



Announcements 4/6

Assignments that are due soon

- Pre-Class Assignment Week 10-W
- Bonus point for Unit 3:
Exam 2 Wrapper (Close 11:59 PM 4/6)

Office Hour

- Dr. Wang's: Tue 10-11 AM at KC 230
- Paige Kwan's: Mon 11:30 AM -12:30 PM at the TLC (Demille 160P)
- Saige Douglass's: Tue 12:00-1:00 PM at the TLC (Demille 160P)



SOUND HEALING

TUNE INTO WELLNESS. VIBRATE WITH HARMONY.

Immerse yourself in therapeutic sounds that calm the mind,
heal the body, and uplift the spirit.

MONDAY, APRIL 6 @ 6:00 PM
FISH INTERFAITH CENTER



Integrated Rate Law

The integrated Rate Law:

- Describes the relationship between concentration and time
- Derived from the Rate Law.



First order reaction:

$$r = k[A] = \frac{-d[A]}{dt} \xrightarrow{\text{Rearrange}} \frac{-d[A]}{[A]} = -d \ln[A] = k dt$$

Integrate $\downarrow$

$$\int_{[A]_0}^{[A]_t} -d \ln[A] = \int_0^t k dt$$

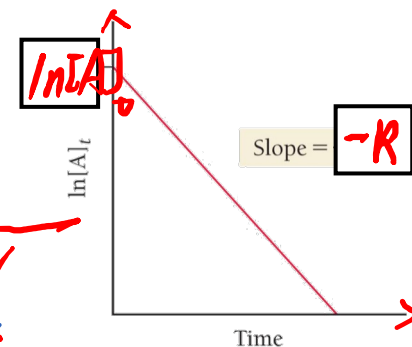
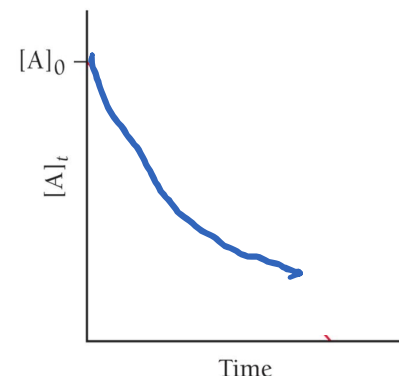
$$-(\ln[A]_t - \ln[A]_0) = kt$$

$$e^{\ln[A]_t} = [A]_t = e^{\ln[A]_0 - kt} = e^{\ln[A]_0} \cdot e^{-kt}$$

- Integrated Rate Law for zero order reaction: $\ln[A]_t = \ln[A]_0 - kt$

① $y = b + ax$

② $[A]_t = [A]_0 \cdot e^{-kt}$



Integrated Rate Law

The integrated Rate Law:

- Describes the relationship between concentration and time
- Derived from the Rate Law.

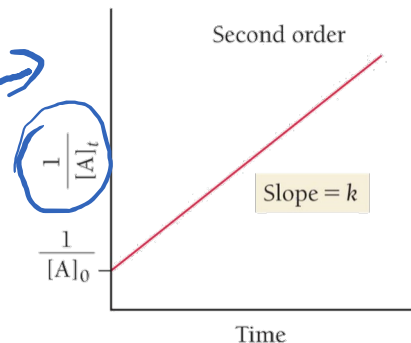
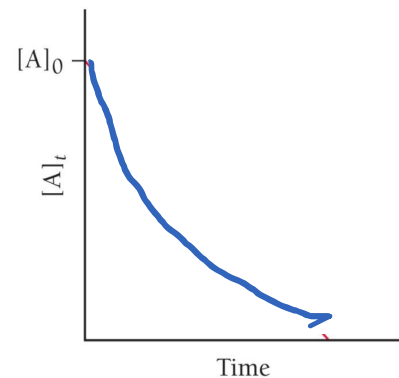


Second order reaction:

$$r = k[A]^2 = \frac{-d[A]}{dt} \xrightarrow{\text{Rearrange}} \frac{-d[A]}{[A]^2} = d\left(\frac{1}{[A]}\right) = kdt$$

$$\text{Integrate} \downarrow \int_{[A]_0}^{[A]_t} d\left(\frac{1}{[A]}\right) = \int_0^t kdt$$

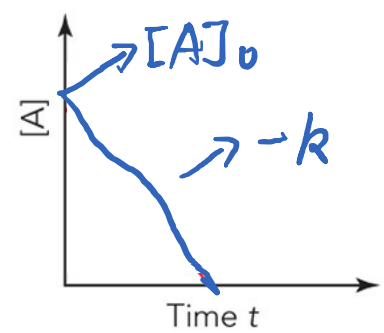
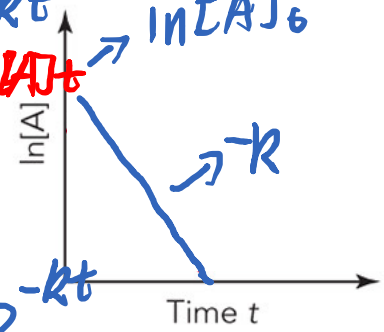
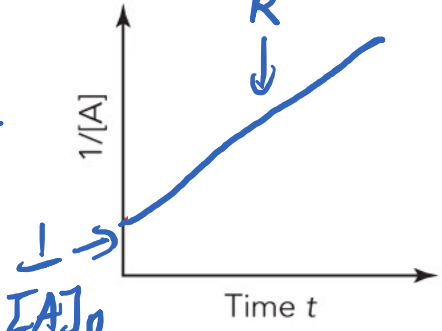
$$\Rightarrow \frac{1}{[A]_t} - \frac{1}{[A]_0} = kt$$



- Integrated Rate Law for zero order reaction: $\frac{1}{[A]_t} = \frac{1}{[A]_0} + kt$

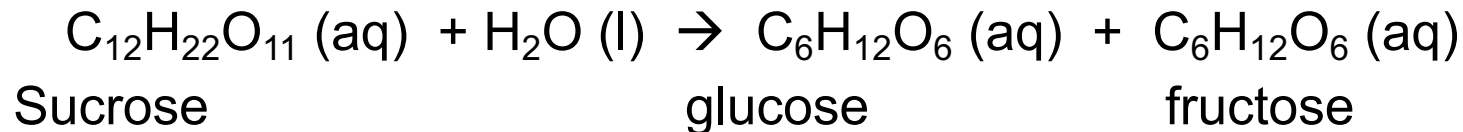
$$y = b + ax$$

Summary of Rate Law and Integrated Rate Law

Order	Rate Law	Units of k	Integrated Rate Law	Straight-Line Plot
0	Rate = k	$\frac{M}{s}$	$[A]_t = [A]_0 - kt$	
1	Rate = $k[A]$	$\frac{1}{s}$	$\textcircled{1} \ln[A]_t = \ln[A]_0 - kt$ $kt = \ln[A]_0 - \ln[A]_t$ $t = \frac{\ln \frac{[A]_0}{[A]_t}}{k}$ $\textcircled{2} [A]_t = [A]_0 \cdot e^{-kt}$	
2	Rate = $k[A]^2$	$\frac{1}{M \cdot s}$	$\frac{1}{[A]_t} = \frac{1}{[A]_0} + kt$	

Your Turn!

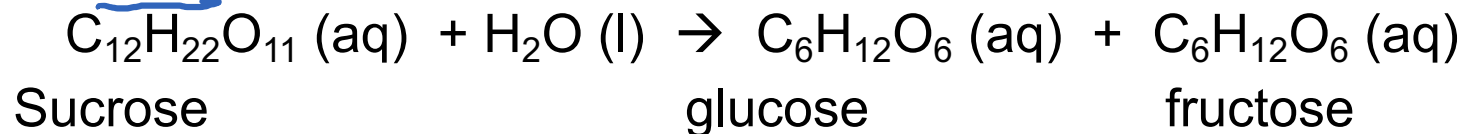
The acidic hydrolysis of sucrose into glucose and fructose is first order with respect to sucrose. The rate constant for this reaction is $6.17 \times 10^{-4} \text{ s}^{-1}$ at 35°C .



- Write the complete Rate Law for the reaction.
- If the initial concentration of sucrose is 0.40 M, what is the initial rate of reaction?

Your Turn!

The acidic hydrolysis of sucrose into glucose and fructose is first order with respect to sucrose. The rate constant for this reaction is $6.17 \times 10^{-4} \text{ s}^{-1}$ at 35°C .



a. Write the complete Rate Law for the reaction.

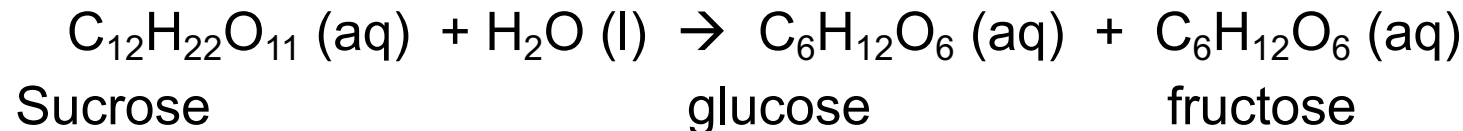
$$r = 6.17 \times 10^{-4} \text{ s}^{-1} [\text{C}_{12}\text{H}_{22}\text{O}_{11}] \quad \leftarrow$$

b. If the initial concentration of sucrose is 0.40 M, what is the initial rate of reaction?

$$r = 6.17 \times 10^{-4} \text{ s}^{-1} \times 0.40 \text{ M} = 2.47 \times 10^{-4} \text{ M/s}$$

Your Turn!

The acidic hydrolysis of sucrose into glucose and fructose is first order with respect to sucrose. The rate constant for this reaction is $6.17 \times 10^{-4} \text{ s}^{-1}$ at 35°C .

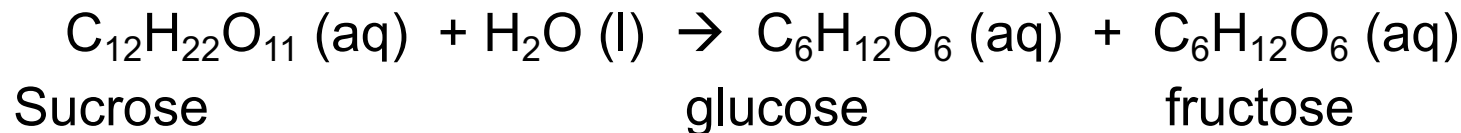


- c. What will be the sucrose concentration after 2 hours, if the initial sucrose concentration is 0.40M?
- d. How many minutes will it take for the sucrose concentration to drop from 0.40 M to 0.30 M?

Your Turn!

$$\ln[A]_t = \ln[A_0] - kt$$

The acidic hydrolysis of sucrose into glucose and fructose is first order with respect to sucrose. The rate constant for this reaction is $6.17 \times 10^{-4} \text{ s}^{-1}$ at 35°C .



- c. What will be the sucrose concentration after 2 hours, if the initial sucrose concentration is 0.40M?

$$\begin{aligned} [A]_t &= [A_0] e^{-kt} = (0.40 \text{ M}) e^{-6.17 \times 10^{-4} \text{ s}^{-1} \times 2 \text{ h} \times \frac{60 \text{ min}}{1 \text{ h}} \times \frac{60 \text{ s}}{1 \text{ min}}} \\ &= 4.71 \times 10^{-3} \text{ M} \end{aligned}$$

- d. How many minutes will it take for the sucrose concentration to drop from 0.40 M to 0.30 M?

$$\begin{aligned} \Rightarrow \ln(0.30 \text{ M}) &= \ln(0.40 \text{ M}) - 6.17 \times 10^{-4} \text{ s}^{-1} \times t \\ t &= 466 \text{ s} \times \frac{1 \text{ min}}{60 \text{ s}} = 7.8 \text{ min} \end{aligned}$$

Half-Life for a reaction

Half-Life ($t_{1/2}$) a reaction: the time required for the concentration of a reactant to fall to one-half of its initial value: $[A]_t = \frac{1}{2}[A]_0$

Zero order reaction:

$$[A]_t = [A]_0 - kt$$



$$[A]_t = \frac{1}{2} [A]_0$$

$$\frac{1}{2} [A]_0 = [A]_0 - kt$$

$$kt = \frac{[A]_0}{2}$$

$$t_{\frac{1}{2}} = \frac{[A]_0}{2k}$$

$$\text{Half life: } t_{1/2} = \frac{[A]_0}{2k}$$

First order reaction:

$$\ln[A]_t = \ln[A]_0 - kt$$



$$[A]_t = \frac{1}{2} [A]_0$$

$$\ln \frac{[A]_0}{2} = \ln[A]_0 - kt$$

$$\cancel{\ln[A]_0} - \ln 2 = \cancel{\ln[A]_0} - kt$$

$$kt = \ln 2$$

$$t_{\frac{1}{2}} = \frac{\ln 2}{k}$$

$$\text{Half life: } t_{1/2} = \frac{\ln 2}{k} = \frac{0.693}{k}$$

Second order reaction:

$$\frac{1}{[A]_t} = \frac{1}{[A]_0} + kt$$



$$[A]_t = \frac{1}{2} [A]_0$$

$$\frac{2}{[A]_0} = \frac{1}{[A]_0} + kt$$

$$kt = \frac{1}{[A]_0}$$

$$t_{\frac{1}{2}} = \frac{1}{k[A]_0}$$

$$\text{Half life: } t_{1/2} = \frac{1}{k[A]_0}$$

Carbon Dating

Radiocarbon dating (also referred to as carbon dating or carbon-14 dating) is a method for determining the age of an object containing organic material.

When the animal or plant dies, the C-14 in it starts to decay following by the reaction:
 $^{14}_6\text{C} \rightarrow ^{14}_7\text{N} + e^- + \bar{\nu}_e$. The decay of C-14 is a first order reaction.

The ^{14}C content of an ancient piece of wood was found to be 1/10 of living trees. How many years old is the piece of wood? ^{14}C : $t_{1/2} = 5730$ years

$$t_{1/2} = \frac{\ln 2}{k} = 5730 \text{ yr.}$$

$$k = \frac{\ln 2}{5730 \text{ yr}} = 1.21 \times 10^{-4} \text{ yr}^{-1}$$

$$t = \frac{\ln \frac{[A]_0}{[A]_t}}{-k} = \frac{\ln \frac{1}{0.1}}{-1.21 \times 10^{-4} \text{ yr}^{-1}} = 19034 \text{ yr}$$



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The ^{14}C content of an ancient piece of wood was found to be 1/10 of living trees. How many years old is the piece of wood? ^{14}C : $t_{1/2} = 5730$ years

$$t_{\frac{1}{2}} = \frac{\ln 2}{k} = 5730 \text{ yr}$$

$$k = 1.21 \times 10^{-4} \text{ yr}^{-1}$$

$$\frac{1}{10} [\cancel{A}]_0 = [\cancel{A}_0] e^{-kt}$$

$$\frac{1}{10} = e^{-kt}$$

$$\ln \frac{1}{10} = -1.21 \times 10^{-4} \text{ yr}^{-1} \times t$$

$$t = 19034 \text{ yr}$$



Announcement 4/8

Assignments that are due soon

- Pre-Class Assignment Week 10-F

Chem RePS Session

Wed 2:00 - 3:30 PM at KC 230

1st Chem RePS or SI worksheets for Unit 3 will be closed on 11:59 PM today

Office Hour

- Dr. Wang's: Thu 10-11 AM on Zoom

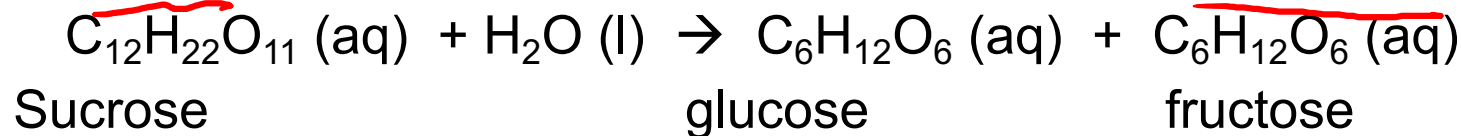
SI Session

Saige Douglass – Wed 5:30 – 7:30 PM at Hashinger 412

Paige Kwan - Thu 5 - 7 PM at KC134

Your Turn!

The acidic hydrolysis of sucrose into glucose and fructose is first order with respect to sucrose. The rate constant for this reaction is $6.17 \times 10^{-4} \text{ s}^{-1}$ at 35°C .



- e. Calculate the half-life for the hydrolysis of sucrose reaction.

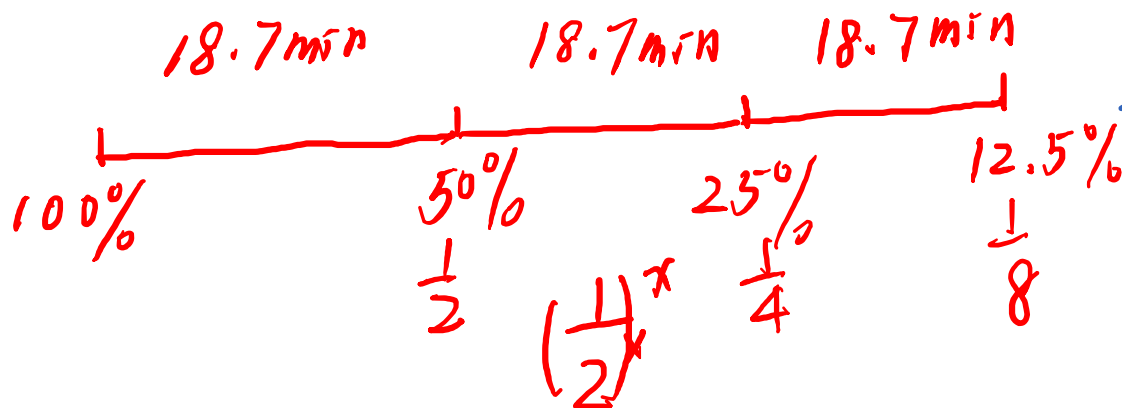
$$t_{1/2} = \frac{\ln 2}{k} = \frac{\ln 2}{6.17 \times 10^{-4} \text{ s}^{-1}} = 1123 \text{ s} \times \frac{1 \text{ min}}{60 \text{ s}} = \underline{18.7 \text{ min}}$$

- f. How many minutes would it take for 75% of the sucrose to react?

$$[A]_t = 25\% [A]_0 \quad \ln \frac{25\% [A]_0}{[A]_0} = -6.17 \times 10^{-4} \text{ s}^{-1} t$$

- g. Solve part (f) by an analysis of half lives

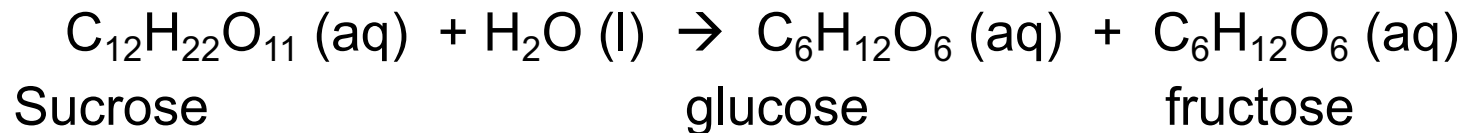
$$t = \frac{\ln 0.25}{-6.17 \times 10^{-4} \text{ s}^{-1}} = 2 \times 1123 \text{ s} = \underline{37.4 \text{ min}}$$



e. 18.7 min; f. 37.4 min

Your Turn!

The acidic hydrolysis of sucrose into glucose and fructose is first order with respect to sucrose. The rate constant for this reaction is $6.17 \times 10^{-4} \text{ s}^{-1}$ at 35°C .



e. Calculate the half-life for the hydrolysis of sucrose reaction.

$$t_{1/2} = \frac{\ln 2}{k} = \frac{\ln 2}{6.17 \times 10^{-4} \text{ s}^{-1}} = 1123 \text{ s} \times \frac{1 \text{ min}}{60 \text{ s}} = 18.7 \text{ min}$$

f. How many minutes would it take for 75% of the sucrose to react?

$$\ln \frac{(1-75\%) [A]_0}{[A]_0} = -6.17 \times 10^{-4} \text{ s}^{-1} t \quad t = 2247 \text{ s} \times \frac{1 \text{ min}}{60 \text{ s}} = 37.4 \text{ min}$$

g. Solve part (f) by an analysis of half lives

$$\text{After } t_{1/2}, 50\% \text{ left; After } 2 \times t_{1/2}, 25\% \text{ left}$$

$$18.7 \text{ min} \times 2 = 37.4 \text{ min}$$

Your Turn!

first order

Iodine, I-131, is used as a metabolic tracer in hospitals. It has a half-life of 8.07 days. How long before the activity falls to 1% of the initial value?

Your Turn!

first

2

Iodine, I-131, is used as a metabolic tracer in hospitals. It has a half-life of 8.07 days. How long before the activity falls to 1% of the initial value?

$$\textcircled{1} \quad t_{\frac{1}{2}} = \frac{\ln 2}{k} = 8.07 \text{ d} \quad k = 0.086 \text{ d}^{-1}$$

$$\ln \left(\frac{1\% [A]_0}{[A]} \right) = -0.086 \text{ d}^{-1} t \quad \leftarrow$$

$$t = 53.6 \text{ days}$$

first order

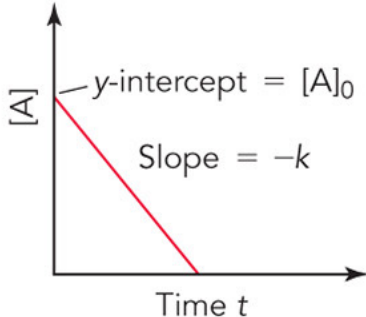
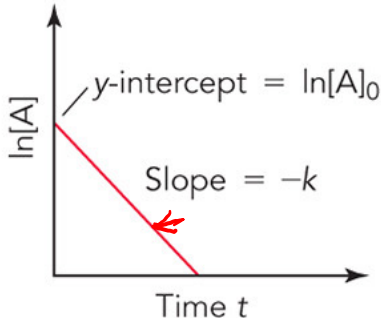
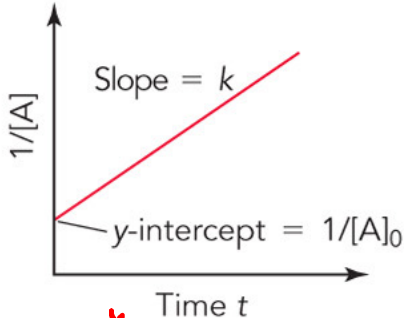


$$0.01 = \left(\frac{1}{2}\right)^x$$

$$x = 6.64$$

$$8.07 \text{ d} \times 6.64 = 53.6 \text{ d.}$$

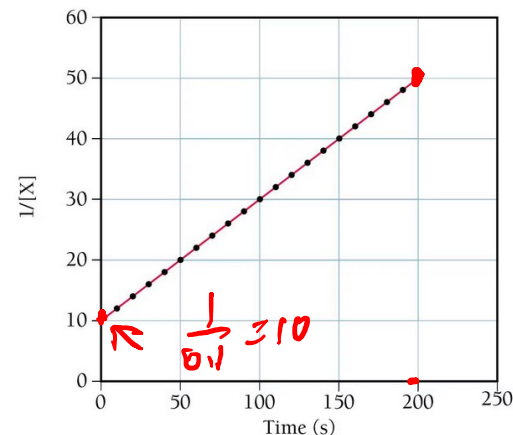
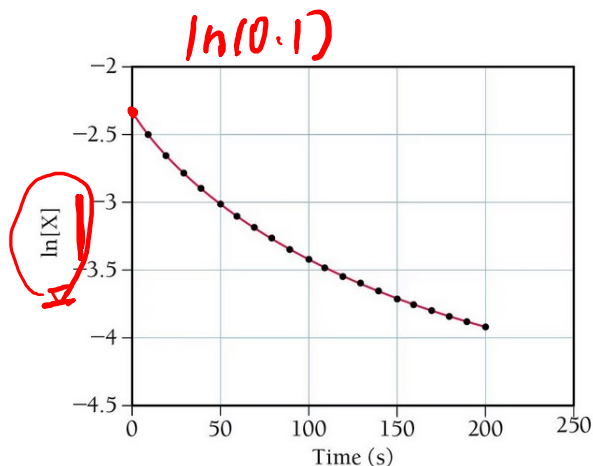
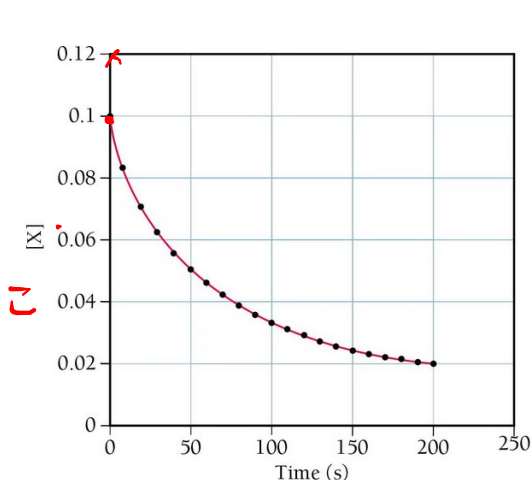
Summary of Rate Law

Order	Rate Law	Units of k	Integrated Rate Law	Straight-Line Plot	Half-Life Expression
0	$\text{Rate} = k[A]^0$	$M \cdot s^{-1}$	$[A]_t = -kt + [A]_0$		$t_{1/2} = \frac{[A]_0}{2k} = \frac{1}{k} \frac{[A]_0}{2}$
1	$\text{Rate} = k[A]^1$	s^{-1}	$\ln[A]_t = -kt + \ln[A]_0$ $\ln \frac{[A]_t}{[A]_0} = -kt$		$t_{1/2} = \frac{0.693}{k} = \frac{1}{k} (0.693)$
2	$\text{Rate} = k[A]^2$	$M^{-1} \cdot s^{-1}$	$\frac{1}{[A]_t} = kt + \frac{1}{[A]_0}$		$t_{1/2} = \frac{1}{k[A]_0} = \frac{1}{k} \frac{1}{[A]_0}$

Your Turn!



The concentration of X for a reaction $X \rightarrow \text{products}$ was measured as a function of time. Examine the graphs below and answer the questions.



- 1) What is the order of the reaction? 2
- 2) What is the value of k for the reaction?

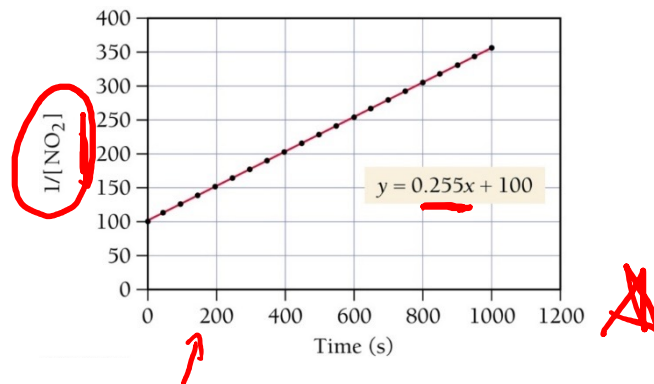
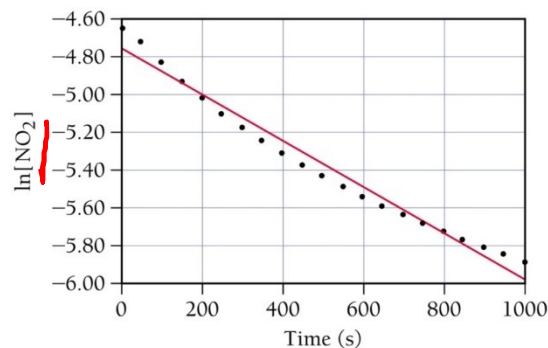
$$k = \frac{\Delta y}{\Delta x} = \frac{50 - 10}{200 - 0} = \frac{40}{200} = 0.2 \text{ M}^{-1} \cdot \text{s}^{-1}$$

- 3) What is the rate law for the reaction? $r = 0.2 \text{ M}^{-1} \cdot \text{s}^{-1} [X]^2$



Your Turn!

For the reaction $\text{NO}_2 (\text{g}) \rightarrow \text{NO} (\text{g}) + \text{O} (\text{g})$, the concentration of NO_2 is monitored at a fixed temperature as a function of time during the decomposition reaction.



1) What is the order of the reaction? Write out the complete rate law.

2 $r = 0.255 \text{ M}^{-1} \text{ s}^{-1} \cdot [\text{NO}_2]^2$

2) What is the rate of the reaction at the $[\text{NO}_2]$ of 0.02 M.

$$r = 0.255 \text{ M}^{-1} \text{ s}^{-1} \cdot (0.02 \text{ M})^2 = 1.02 \times 10^{-4} \frac{\text{M}}{\text{s}}$$

3) Calculate the time, in s, when $[\text{NO}_2]$ degrades from initial concentration of 0.02 M to 0.01 M.

$$t_{\frac{1}{2}} = \frac{1}{0.255 \text{ M}^{-1} \text{ s}^{-1} \times 0.02 \text{ M}} = 196 \text{ s}$$

$$\frac{1}{0.01} = 0.255 t + \frac{1}{0.02}$$

196 s

2) $1.02 \times 10^{-4} \text{ M/s}; 3) 196 \text{ s}$